Campaign Performance Summary Report

Analyst:

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Data Analyst | Data-driven insights for smarter marketing decisions

Business Objective

The goal of this analysis was to identify the most cost-efficient and profitable marketing campaign for a retail startup facing funding constraints.
With limited resources, the company needed to focus on a single campaign that delivers maximum customer acquisition and profitability while minimizing marketing costs.

Methodology

1 Data Cleaning & Preparation

- Imported and cleaned historical order data using Python (Pandas).
- Standardized inconsistent values (removed special symbols, corrected formats, and handled missing data).
- Created new computed metrics:
    - Revenue = ProductPrice × Quantity
    - Cost = ProductCost × Quantity
    - Margin = Revenue − Cost
- Removed duplicates and invalid dates to ensure data accuracy.

2 Campaign Performance Metrics

Aggregated data by AcquisitionSource to measure:

- Number of unique orders and customers
- Total revenue, cost, and profit margin
- Average Order Value (AOV) and Margin %

3 Advanced Analysis

- Added hypothetical marketing spend per campaign to calculate Customer Acquisition Cost (CAC).
  (CAC = Spend ÷ Customers)
- Created a Budget vs Predicted Customers model using Linear Regression to simulate how customer numbers could increase with higher budgets.

4 Visualization

Used Matplotlib to design:

- Bar Chart: CAC by Campaign
- Line Chart: Weekly Revenue Trends
- Scatter Plot: Budget vs Predicted Customers

Dataset Overview

- Source: Order_Data.csv
- Size: 13 columns × 55912 rows
- Key features:
    - Acquisition Source: campaign/channel (Google Ads, Meta Ads, YouTube, etc.)
    - ProductPrice, ProductCost, OrderQuantity, for revenue and cost calculations
    - CustID, OrderID, for unique customer and order counts
    - Fraud, fraud flag (all 0% in dataset)

Metrics Computed

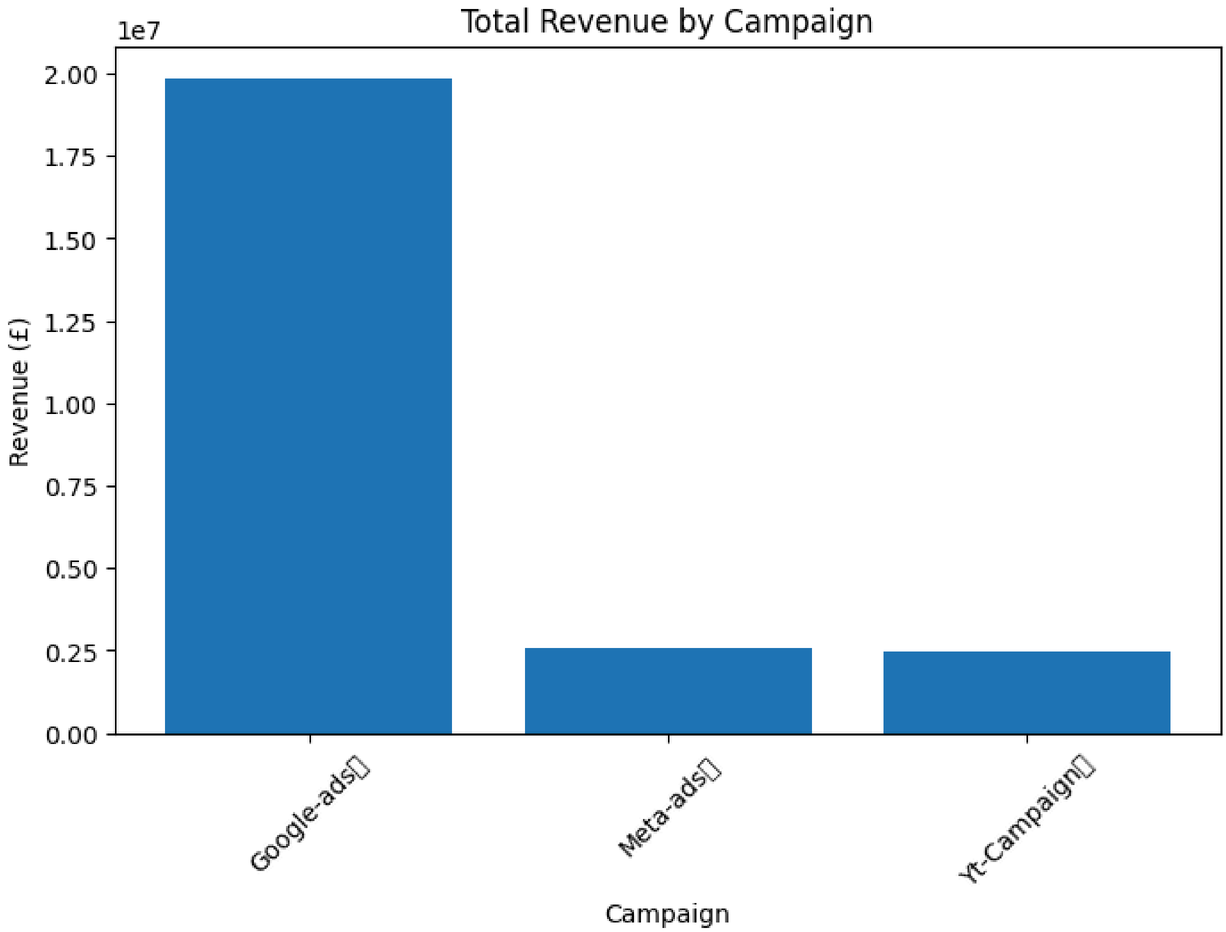
KPI	Formula	Meaning
Revenue	ProductPrice × OrderQuantity	Total sales value
Cost	ProductCost × OrderQuantity	Total cost incurred
Margin	Revenue − Cost	Profit earned
AOV (Average Order Value)	Revenue ÷ Orders	Average sale per customer
Margin %	Margin ÷ Revenue	Profit efficiency
Fraud Rate	Fraud Orders ÷ Total Orders	% of suspicious transactions

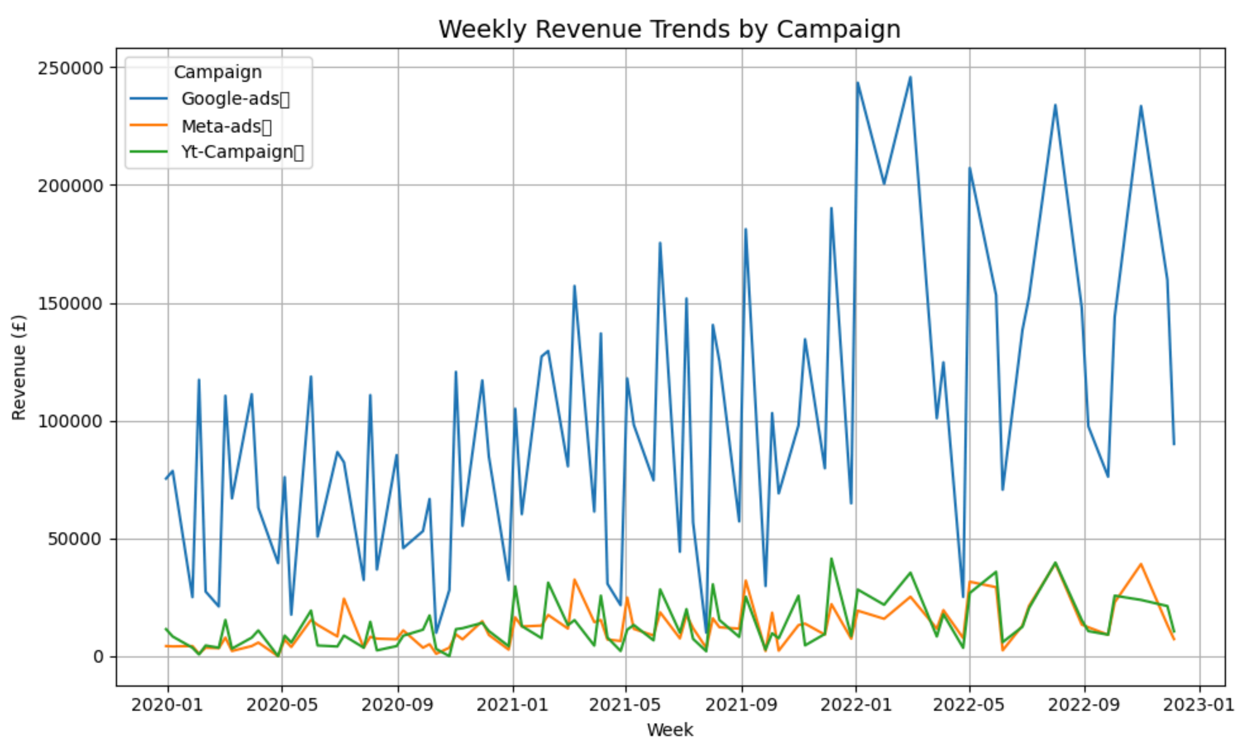
Results Summary

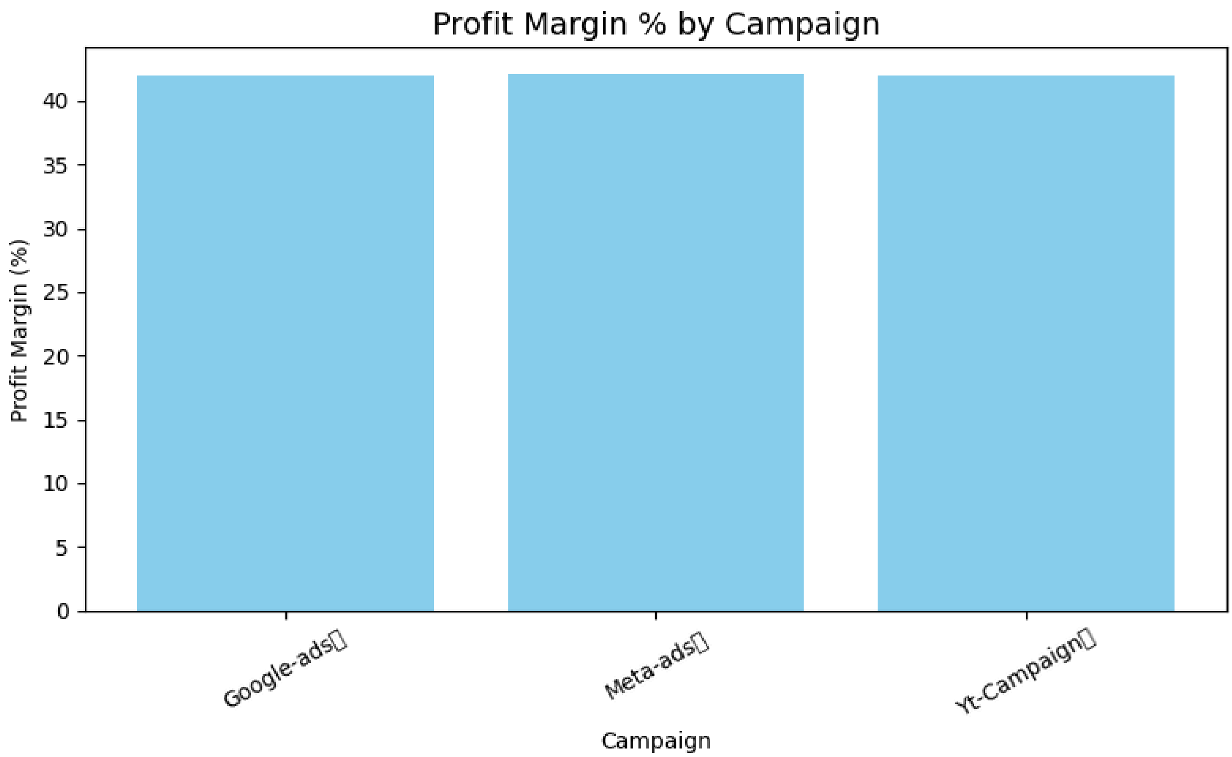
Campaign	Orders	Customers	Revenue (£)	Cost (£)	Margin (£)	Margin %	Fraud Rate	Score
Google Ads	Highest	Highest	Lower than others	Moderate	Highest	Top performer	0%	1.00 (best)

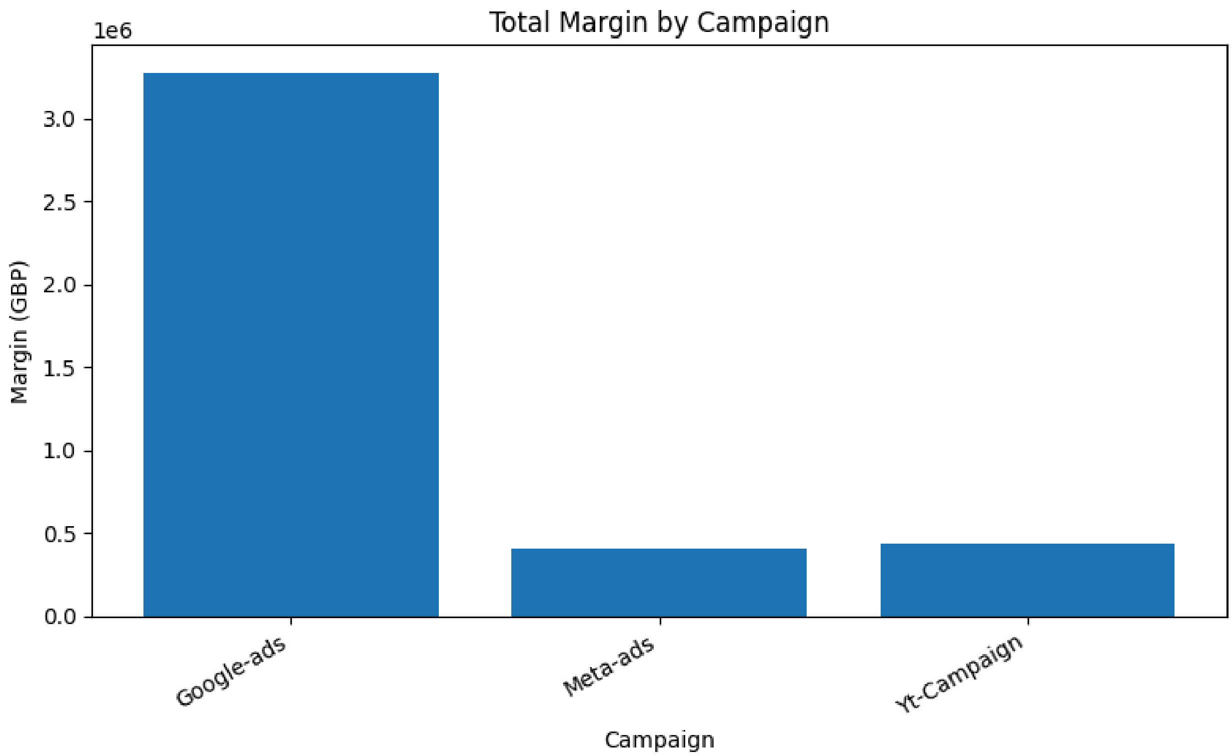
Meta Ads	Medium	Medium	Higher revenue	High cost	Lower margin	Mid performer	0%	0.76
YouTube Campaign	Moderate	Moderate	Medium	Medium	Mid margin	Stable	0%	0.72

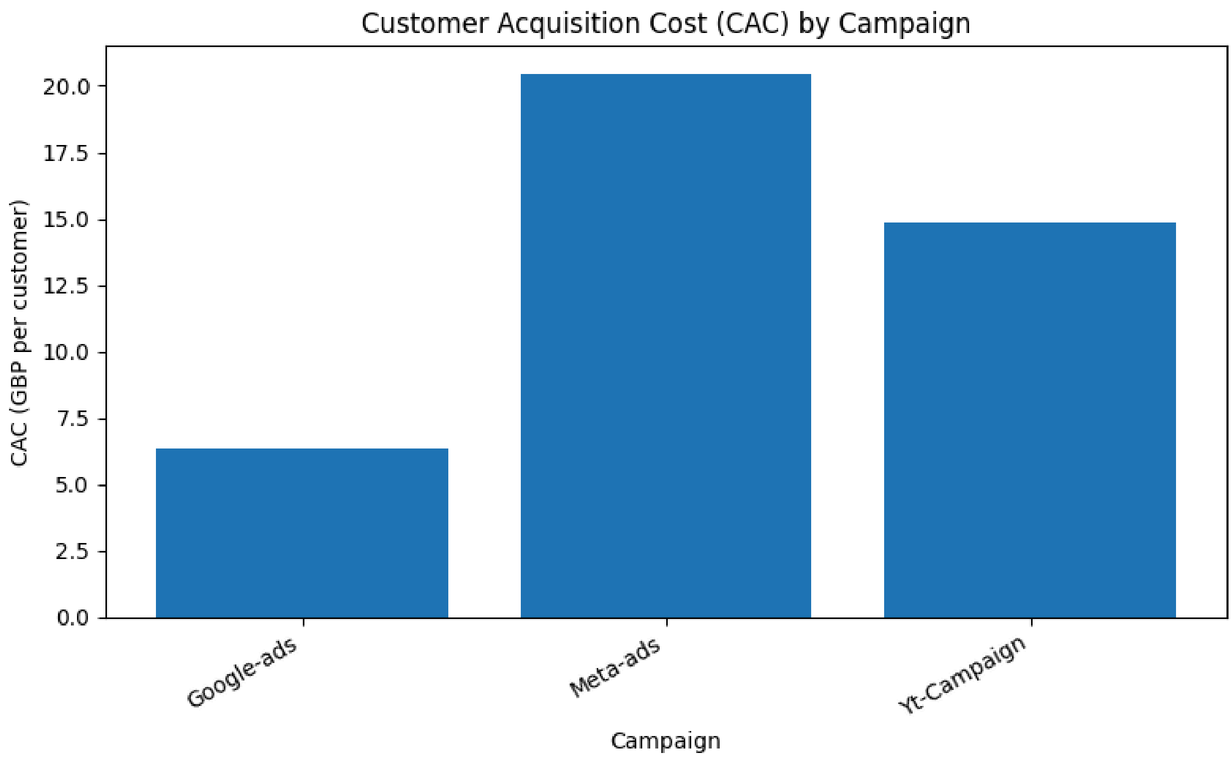
Campaign	Customers	CAC (£)	Profit Margin (£)
Google-Ads	7,881	**Lowest**	Highest
Meta-Ads	1,954	**Highest**	Moderate
YT-Campaign	2,017	Medium	Stable

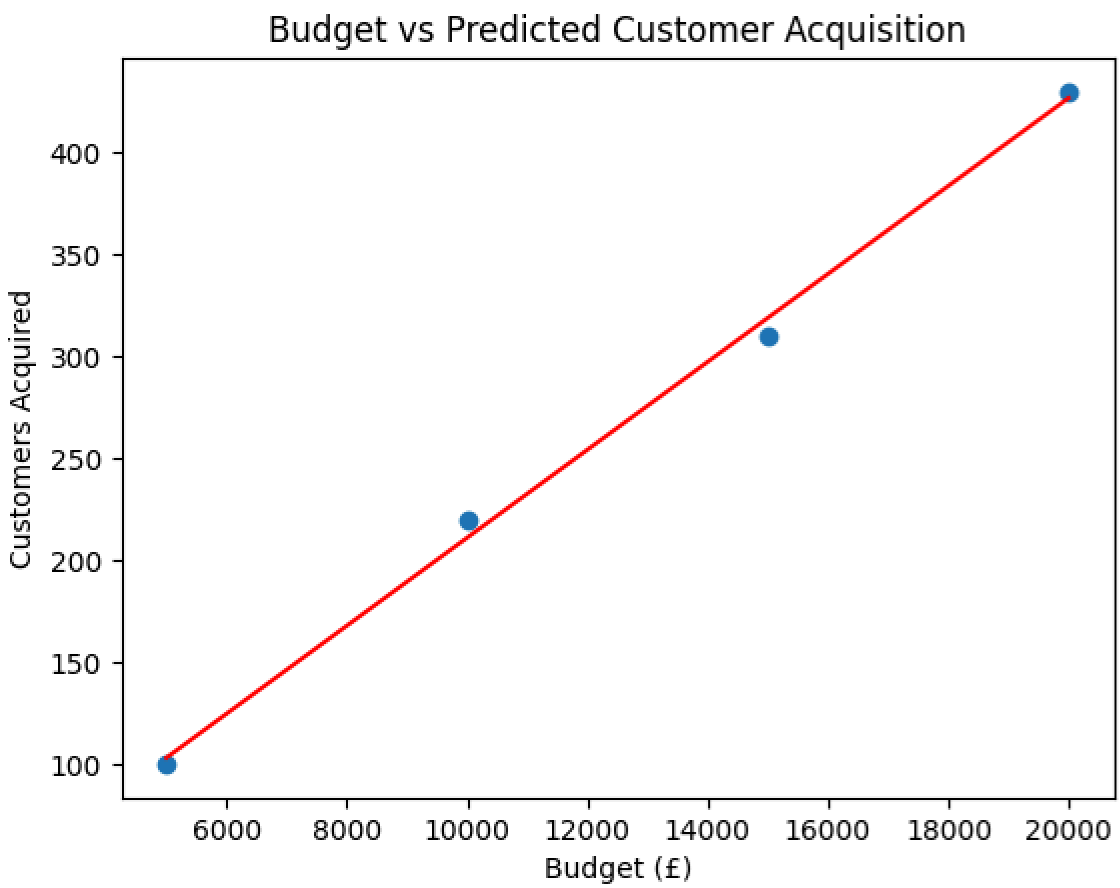

Total Revenue by Campaign
1e7
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Revenue (£)
Google-ads
Meta-ads
Yt-Campaign
Campaign


Weekly Revenue Trends by Campaign
Campaign
Google-ads
Meta-ads
Yt-Campaign
250000
200000
150000
100000
50000
0
Revenue (£)
2020-01
2020-05
2020-09
2021-01
2021-05
2021-09
2022-01
2022-05
2022-09
2023-01
Week


Profit Margin % by Campaign
Profit Margin (%)
0
5
10
15
20
25
30
35
40
Google-ads□
Meta-ads□
Yt-Campaign□
Campaign


Total Margin by Campaign
1e6
Margin (GBP)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Google-ads
Meta-ads
Yt-Campaign
Campaign


Customer Acquisition Cost (CAC) by Campaign
CAC (GBP per customer)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Google-ads
Meta-ads
Yt-Campaign
Campaign


Budget vs Predicted Customer Acquisition
Customers Acquired
100
150
200
250
300
350
400
6000
8000
10000
12000
14000
16000
18000
20000
Budget (£)

Key Insights and Interpretation

1. Google Ads campaigns consistently outperformed others, achieving:
    - The lowest CAC,
    - The highest customer acquisition volume, and
    - The best overall profit margin.
2. Meta-Ads campaign showed the highest CAC, meaning it cost the company significantly more to acquire each customer, reducing profitability.
3. YT-Campaign maintained moderate CAC and profit levels, performing fairly well but not as efficiently as Google Ads.
4. The Budget vs Predicted Customers analysis confirmed a positive correlation, as budget increases, the number of customers acquired also increases.

This supports the recommendation to strategically increase investment in the most efficient campaign (Google-Ads).

Recommendations

1. Continue with Google-Ads as the main acquisition channel.
    - Allocate a higher portion of the marketing budget to Google-Ads for optimal ROI.
    - Maintain fraud monitoring and conversion tracking.
2. Reduce or pause spending on Meta-Ads until CAC efficiency improves through A/B testing or new targeting strategies.
3. Optimize YT-Campaign by experimenting with ad creatives and keywords to lower CAC and boost conversions.
4. Next Steps:
    - Integrate ad spend data for real CAC validation.
    - Track weekly AOV, CAC, and ROI to measure improvement.
    - Run an A/B test comparing landing pages for Google and Meta campaigns.

Visual Insights

- Bar Chart: Google-Ads had the lowest CAC, while Meta-Ads was the most expensive.
- Line Chart: Weekly revenue trends showed consistent performance from Google-Ads.
- Regression Chart: Demonstrated that increasing budget directly led to higher customer acquisition.

Tools Used

- Python (Pandas, NumPy, Matplotlib) – Data cleaning, analysis, and visualization
- Google Colab – Execution environment
- GitHub – Version control and project publishing

Conclusion

This analysis clearly shows that Google-Ads is the most cost-effective and profitable campaign for the business.

Focusing marketing spend on this channel will help the company maximize customer acquisition, reduce acquisition costs, and improve profitability, ensuring long-term financial stability.